

The OpenAI Navier–Stokes and Euler Blow-Up Proofs: A Physical Reading, Not a Physical Refutation

The MechanicaFluidorum Program
SocrateAI Lab

September 2026

Abstract

In September 2026, an OpenAI multi-agent system produced a Lean 4-verified proof of finite-time blow-up for the forced 3D Navier–Stokes equations (Millennium Prize Alternatives C and D) and for the unforced Euler equations. The proof is syntactically flawless: zero **sorrys**, zero custom axioms, no weakened norms. The Clay Millennium Prize problems, Navier–Stokes existence and smoothness included, are precise questions about a specific mathematical model; they were never framed, and are not understood by working mathematicians, as claims about how real fluids behave — the model is already known to break down under compressibility, cavitation, molecular discreteness, and other effects well before any putative singularity. Read that way, OpenAI’s result is exactly what it claims to be: a rigorous statement about the continuum incompressible Navier–Stokes system on $\mathbb{R}^3$.

Our contribution is not a refutation but a physical reading. Because the collapsing core keeps a radial Reynolds number of order one, its scales are diffusive, $\ell_r \simeq \sqrt{\nu t}$ and $u \simeq \sqrt{\nu/t}$ up to factors $t^{\pm h}$ with $h < 1/100$, so the reading is essentially independent of the initial vortex size and of the choice of units. Compressibility, rarefaction and viscous heating then become order-one effects at a single length $\ell_* = \nu/c_s$ (0.7 nm in water, 45 nm in air), reached a few picoseconds (water) or nanoseconds (air) before the mathematical singularity: the three validity limits are one limit, and it can be written as a local vorticity bound $|\omega| \lesssim c_s^2/\nu$. For liquids at ambient pressure a fourth limit, cavitation, is reached three decades earlier, when the core is still ~ 70 nm wide. We also show that (i) intensive local energy density and enstrophy diverge ($\sim \tau^{-1.010}$, $\sim \tau^{-0.515}$) while the Prize’s global L^2 bound is respected; (ii) the apparent 10^{28} condition number of the proof’s moment-matching system is a non-dimensionalization artifact ($\kappa \approx 4.1 \times 10^5$ once scaled); (iii) the force is specified by working backward from the singularity, a legitimate existence-proof technique that leaves it without independent physical origin; and (iv) the Euler datum requires coherent oscillatory structure at scales below any molecular cutoff, where the missing physics is viscosity itself. None of this makes the Lean proofs wrong, and none of it proves anything about turbulence in real fluids — both points made forcefully by the fluid-dynamics community when an early version of this audit was shared publicly (§10). We close with the research directions raised in that discussion, a neuro-symbolic-physical architecture for keeping AI-generated proofs labelled as mathematical or physical results, and one clearly-labelled hypothesis on cutoff regularizations.

1 Introduction

The application of large-scale AI agents to formalize mathematics in Lean 4 guarantees logical consistency: if the proof compiles, the chain $A \implies B$ is syntactically flawless. On September 8, 2026, OpenAI announced that an internal multi-agent system had produced such a proof of finite-time blow-up for the forced 3D Navier–Stokes equations and, separately, for the unforced Euler equations [2, 4, 3]. (Figures for the size of that system — “10,000 agents” — circulated

in secondary coverage but appear in neither paper; we do not repeat them as fact.) The announcement was accompanied by a public discussion of priority with other groups working on the Euler problem. We do not adjudicate it, and it plays no role in what follows, except as a reminder that a result produced under competitive pressure, however impeccably kernel-checked, still benefits from independent, unhurried scrutiny of what it actually establishes — which is the purpose of this paper.

The result does not arrive in a vacuum. Finite-time blow-up was known for an averaged Navier–Stokes model [10], for $C^{1,\alpha}$ Euler data [23], and, with computer assistance, for smooth Euler and Boussinesq data in domains with boundary [24] following numerical evidence [22]; non-uniqueness of Leray solutions of the *forced* Navier–Stokes equations [26] had already shown how much a well-chosen force can do. What is new is smooth, compactly supported data on $\mathbb{R}^3$ (Euler) and a smooth force acting on a fluid at rest (Navier–Stokes), with kernel-checked proofs.

Our purpose is not to contest the mathematics. A proof mapping $A \implies B$ is syntactically true regardless of how physically realistic A is, and the Millennium Prize problems were never framed as claims about physical fluids: Fefferman’s statement [1] asks precise questions about a specific continuum model, chosen because it is tractable and physically motivated, not because it is physically exact. This is also the reading offered independently, and largely spontaneously, by the fluid-dynamics community when an early version of this audit was shared for discussion on `r/FluidMechanics` (quoted with usernames as posted publicly; see §10 for the full exchange) [6]: commenters repeatedly made the point that Navier–Stokes “has always been a useful approximation and not fully physical,” that a blow-up result mainly tells us where a specific mathematical model runs out of validity, and that the more interesting question is what a genuinely physical admissibility criterion — analogous to the entropy condition used to select physical weak solutions of the inviscid Burgers equation — might look like.

Since the Lean 4 kernel forecloses algebraic error, this paper’s contribution is not a bug hunt in that sense. It is instead a quantitative account of how far OpenAI’s specific constructed solution travels from the physical regime the Navier–Stokes model is normally used to describe, spanning formal verification, arbitrary-precision numerical analysis, and thermodynamic first principles — followed by an outlook (§10) that takes the community’s framing seriously: we propose concrete physics-integration research directions, a neuro-symbolic-physical architecture for grounding future AI proof search in physical admissibility, and flag one untested, string-theory-motivated hypothesis for further investigation.

2 The Syntactic Health: Lean 4 Purity and Gevrey Cutoffs

2.1 AST Verification Results

An automated crawl of the Abstract Syntax Tree confirms the proof is structurally flawless:

There are no opaque axioms, no weakened Sobolev norms, and the global L^2 kinetic energy is uniformly bounded. The AI did not “cheat.”

2.2 The Gevrey Bypass: Satisfying C^∞ Without Real-Analyticity

The Millennium Prize requires a perfectly smooth (C^∞) external force. Were the force required to be real-analytic (C^ω), the construction would be impossible outright: a function that is real-analytic and vanishes outside a bounded region must be identically zero everywhere (the identity theorem), so no nontrivial compactly supported force could exist at all. The formalization sidesteps this rigidity with Gevrey-class cutoffs of the form $\chi(q) \sim \exp(-1/q^2)$, which are C^∞ but not real-analytic.

By classical asymptotic analysis, $\exp(-1/q^k)$ has Gevrey index $s = 1 + 1/k$; for $k = 2$ this gives $s = 1.5$ (finite-sample `SymPy` fits with $N \leq 15$ show an empirical pre-asymptotic slope near 2.22, reflecting the fact that the true asymptotic regime is only reached for larger

Criterion	Expected	Found
sorry / admit in core proof	0	0
Custom axiom declarations	0	0
opaque / constant declarations	0	0
Force smoothness type	ContDiff $\mathbb{R} \infty$	ContDiff $\mathbb{R} \infty$
Force support	Compact Positive Time	Compact Positive Time
Energy bound	$L_t^\infty L_x^2$ (uniform)	$\exists E, \forall t, \text{kineticEnergy } u \ t \leq E$
Sobolev weakening detected	None	None

Table 1: Lean 4 formal verification results. `sorry` instances exist in `ComparatorChallenges/`, a separate module containing intentional open problems—not part of the core proof.

N). This specific cutoff therefore lies in $G^{1.5} \setminus C^\omega$: infinitely differentiable, with all flat limits $\lim_{q \rightarrow 0^+} d^N[q^{-A} \exp(-1/q^2)]/dq^N = 0$ holding for every N and A (confirmed symbolically), yet not real-analytic. That gap between C^∞ and C^ω is exactly what a compactly supported, nontrivial smooth force requires — there is nothing improper or exotic about invoking it; every physically standard “bump function” construction relies on the same gap. What is worth flagging is not that the cutoff is Gevrey rather than analytic (that is necessary, not suspicious), but how the resulting velocity profile behaves physically once combined with the rest of the construction, which is the subject of §4–§7.

2.3 The Proof Architecture: Internal Reynolds-Stress Cancellation

An earlier draft of this audit mischaracterized the proof’s logical structure as a simple Method of Manufactured Solutions (MMS). In standard MMS, one merely picks an arbitrary smooth velocity field u and defines $f = \partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p$, which is automatically smooth because u was smooth to begin with.

The OpenAI proof is categorically more sophisticated. The paper itself acknowledges [2]:

“For any incompressible flow u and pressure p , we can always define the external force f to be the residual in (1.1). The Navier-Stokes equations then hold by construction.”

But the constructed u is a singular collapsing vortex column. Its raw residual is *not* smooth— $f_{\text{raw}} \rightarrow \infty$ as $\tau \rightarrow 0$. The proof’s core intellectual contribution, spanning Sections 6–9 of [2], is the construction of oscillatory wave packets whose *internal nonlinear Reynolds stresses* $-\nabla \cdot (u' \otimes u')$ precisely cancel the divergent residual of the collapsing core. It is these *internal* stress interactions—not the external force—that tame the singularity. Only after this Reynolds-stress cancellation homogenizes the residual is the remaining smooth remainder declared as $f \in C_c^\infty$.

The Lean 4 code confirms this final assignment: `force := SpacetimeGluing.smoothExtension 1 (tracedResidual u p L) ...`, where `tracedResidual` encodes the *post-cancellation* smooth residual. The docstring reads: “No force is an input to this definition.” The force does not *drive* the singularity; it is the mathematical artifact *left over* after the singularity has been engineered by internal wave dynamics.

3 Matrix Scaling & Non-Dimensionalization of the Moment System

To seamlessly stitch the singular inner vortex to the smooth outer heat flow, the formalization matches five cumulative radial moments via a 5×5 linear system, which we write with matrix A (Proposition B.8 and the moment conditions (5.10)–(5.11) of [2]; an earlier draft of this audit cited “Lemma 8.7”, which is a different lemma). A naive SVD of this unscaled 5×5 dimensional matrix yields a condition number scaling as $\kappa(A) \sim X_R^{7.75}$, which exceeds 10^{28} as $X_R \rightarrow 1000$.

However, rigorous non-dimensionalization analysis reveals that this condition number explosion is a classical numerical linear algebra artifact. The entries of A integrate polynomial basis functions $(x \cdot X_R)^{2i}$ and $(x \cdot X_R)^{2j}$, spanning physical dimensions from X_R^0 up to X_R^8 . Mathematically, the matrix factors as:

$$A = D B D, \tag{1}$$

where $D = \text{diag}(1, X_R^2, X_R^4)$ for the azimuthal block (and $D_z = \text{diag}(X_R, X_R^3)$ for the axial block), and B is a non-dimensional matrix composed of integrals of dimensionless coordinate $x^{2(i+j)}$.

X_R	$\kappa(A)$ (Raw Unscaled)	$\kappa(B)$ (Non-Dimensionalized)
1	4.11×10^5	4.11×10^5
10	2.36×10^{12}	4.11×10^5
100	2.36×10^{20}	4.11×10^5
1000	1.78×10^{28}	4.11×10^5

Table 2: Condition number comparison of raw dimensional matrix A versus non-dimensionalized matrix $B = D^{-1}AD^{-1}$. Once properly scaled, $\kappa(B) \approx 4.11 \times 10^5$ remains completely bounded and independent of X_R .

Applying the diagonal preconditioning matrix D^{-1} removes the X_R dependence entirely. The non-dimensionalized condition number $\kappa(B) \approx 4.11 \times 10^5$ is well-conditioned and uniform for all X_R . Consequently, the moment-matching system is not a source of fragility. Two further remarks are in order. Values of κ above $\sim 10^{16}$ computed in double precision are not reliable in any case; and the condition number of a linear system used to *choose parameters in a proof* is not a dynamical instability of the flow — an earlier draft of this audit conflated the two. The rigorous version of “how fine-tuned is this singularity” is the codimension question of §10.

4 The Thermodynamic Picture: Divergent Local Quantities

Alternative C of the Millennium Prize requires only that total kinetic energy remains uniformly bounded. The authors achieve this by ensuring the volume of the collapsing vortex core shrinks faster than the velocity-squared diverges. This is a scaling feature, not a thermodynamic one, and it is exactly what the Prize permits.

With velocity scales $u_\theta, u_z \sim \tau^{-1/2-h}$ ($h < 1/100$), radial scale $\ell_r \sim \tau^{1/2}$, and axial scale $\ell_z \sim \tau^{1/2-h}$, a SymPy asymptotic analysis yields the scaling exponents in Table 3. Here τ denotes the dimensionless self-similar time-to-blow-up parameter used throughout the OpenAI construction (not, in this table, a value in seconds — see §5 for the physical time conversion).

The viscous dissipation rate is $\varepsilon(t) = \nu \int_{\mathbb{R}^3} |\nabla \times u|^2 dx \sim \tau^{-0.515}$. The Prize requires only that total dissipated energy remain finite, and indeed $\int_0^1 \tau^{-0.515} d\tau < \infty$ —so this constraint is satisfied. But this mathematical concession obscures the physical picture: the *instantaneous* dissipation rate ($\sim \tau^{-0.515}$) and the *intensive local energy density* ($e_{\text{local}} \sim \tau^{-1.010}$) both diverge

Quantity	Exponent (τ^x)	Behavior ($\tau \rightarrow 0$)
Global Kinetic Energy E	+0.4850	$\rightarrow 0$ ✓
Intensive Local Energy Density e_{local}	−1.0100	$\rightarrow \infty$ ✗
Enstrophy $\Omega(t)$	−0.5150	$\rightarrow \infty$ ✗
L^3 norm $\ u\ _3$	−0.0067	$\rightarrow \infty$ ✗
$H^{3/2}$ Sobolev norm $\ u\ _{H^{3/2}}$	−0.5075	$\rightarrow \infty$ ✗
Local temperature rise estimate ΔT	−1.0100	$\rightarrow \infty$ ✗

Table 3: Asymptotic scaling of key physical quantities for the OpenAI blow-up construction ($h = 1/200$). The global L^2 energy—the sole quantity mandated by the Millennium Prize—is the only quantity that remains bounded. Intensive local energy density $e_{\text{local}} = \frac{1}{2}\rho|u|^2 \sim \tau^{-1-2h}$ scales as $\tau^{-1.010}$, while the Sobolev $H^{3/2}$ norm scales as $\tau^{-0.5075}$. ΔT is an external estimate (§4), not a quantity solved for within incompressible Navier–Stokes.

without bound, and the *local* enstrophy density $|\omega|^2 \sim \tau^{-2-2h} = \tau^{-2.010}$ diverges much faster than its spatial integral. It is this divergence of local dissipation density—not the bounded integral—that controls the heating estimate of §5.

Incompressible Navier–Stokes carries no energy or temperature equation of its own: viscous dissipation is simply removed from the kinetic-energy budget, on the implicit assumption that the resulting heating is negligible (formally, that the relevant Eckert number stays small). That assumption is external to the model and has to be checked separately, not derived from it. Diverging local enstrophy means a diverging local viscous heating *rate*; §5 makes this quantitative for a specific fluid (water) and shows that the resulting heating estimate, together with the velocity itself, invalidates the small-Eckert-number and incompressibility assumptions well before the mathematical singularity. This point was raised independently in community discussion of an early version of this audit (§10), where a commenter correctly noted that the thermal argument is an estimate built on top of the incompressible model from outside, not a consequence derivable within it — we agree, and flag deriving it properly (coupling an actual energy/temperature equation, or a cavitation/phase-change model) as an open direction in §10.

The contrast with turbulence is worth stating without a figure. In physical turbulence the Kolmogorov cascade [14] ends at the microscale $\eta = (\nu^3/\varepsilon)^{1/4}$, which in every laboratory or geophysical flow exceeds the molecular length by orders of magnitude; the construction instead concentrates a vanishing fraction of the energy at a scale $\ell_r \rightarrow 0$ with no dissipative floor. (An earlier draft of this audit included a figure comparing a JHTDB isotropic-turbulence spectrum with an “OpenAI pre-singularity spectrum”; the latter was a schematic drawn by hand, not a computed spectrum — no numerical realization of the construction exists — and the figure has been removed; see §8.)

The L^3 norm also diverges, but only marginally ($\|u\|_{L^3} \sim \tau^{-4h/3} = \tau^{-0.0067}$ for $h = 1/200$): the Escauriaza-Seregin-Šverák criterion [7] proves a singularity *cannot* form while $\|u\|_{L^3}$ stays bounded, so this divergence — however small its exponent — is not a violation of anything; it is a necessary, and here barely satisfied, condition for the blow-up to be admissible at all. That the construction sits so close to the critical Serrin threshold is itself informative: it shows how little room the similarity ansatz leaves, and how sensitive the construction is to the precise value of h .

5 Where the Continuum Model Ends: One Scale, Four Limits

5.1 Unit-free core scales

The core of the construction keeps a radial Reynolds number of order one: with $\ell_r \asymp \tau^{1/2}$, $u \asymp \tau^{-1/2-h}$ and radial diffusion $\nu/\ell_r^2 \asymp \tau^{-1}$ in the leading balance [2, §2.1], the proof is written

at viscosity one and τ is the dimensionless time-to-blow-up. Restoring units with an initial core radius ℓ_0 introduces the viscous time unit $T = \ell_0^2/\nu$ and the physical time-to-blow-up $t = T\tau$, and gives

$$\ell_r \simeq \sqrt{\nu t}, \quad u \simeq \sqrt{\nu/t} \left(\frac{\ell_0^2}{\nu t} \right)^h, \quad 0 < h < \frac{1}{100}, \quad (2)$$

with order-one prefactors that the paper does not make explicit. The initial size enters only through the factor $(\ell_0^2/\nu t)^h$, which for $h = 1/200$ lies between 1.2 and 1.3 over the whole range of interest and changes by a factor 1.3 when ℓ_0 changes by 10^6 . *The physical reading below is therefore essentially independent of how large the vortex was to begin with, and of the choice of units.* Earlier drafts of this audit reported numbers for $\ell_0 = 1$ cm ($T = 100$ s) and, at one point, read the dimensionless τ directly as seconds; Table 5 keeps the $\ell_0 = 1$ cm convention for continuity and reports $t = T\tau$.

5.2 Three validity limits are one

Proposition 5.1 (Scaling estimates, $O(1)$ prefactors). *Let c_s be the sound speed, c_p the specific heat, λ the molecular length (mean free path in a gas, intermolecular spacing in a liquid), and neglect the factor $(\ell_0^2/\nu t)^h$. Along the core:*

1. Compressibility. $\text{Ma} = u/c_s$ reaches a prescribed value at $t_{\text{Ma}} \simeq \nu/(\text{Ma}^2 c_s^2)$, when $\ell_r \simeq \nu/(\text{Ma} c_s)$.
2. Rarefaction. $\text{Kn} = \lambda/\ell_r$ reaches a prescribed value at $t_{\text{Kn}} \simeq \lambda^2/(\text{Kn}^2 \nu)$.
3. Viscous heating. The dissipation per unit mass is $\varepsilon \simeq \nu(u/\ell_r)^2$; since $\ell_r^2 \simeq \nu t$, its action over the remaining time t gives $\Delta T \simeq \varepsilon t/c_p \simeq u^2/c_p$, i.e. the Eckert number of the core is of order one.

For a gas, kinetic theory gives $\nu \simeq \frac{1}{2}\bar{c}\lambda$ with $\bar{c} \sim c_s$, so that $\text{Ma} \simeq \text{Kn} \cdot \text{Re}_r$ (von Kármán's relation); with $\text{Re}_r = O(1)$ all three effects become order one at the same length and time,

$$\ell_* \simeq \nu/c_s, \quad t_* \simeq \nu/c_s^2. \quad (3)$$

Equation (3) is the one-line physical content of this audit. In water, $\ell_* \approx 0.67$ nm (about two molecular diameters) and $t_* \approx 0.44$ ps; in air, $\ell_* \approx 45$ nm (essentially the mean free path) and $t_* \approx 0.13$ ns. The statement is generic: it uses only $\text{Re}_r = O(1)$, i.e. Leray-type self-similar scaling, and would apply to any Navier–Stokes singularity of this type, not only to OpenAI's construction. Table 4 evaluates the conventional thresholds ($\text{Ma} = 0.3$, where density variations $\Delta\rho/\rho \approx \text{Ma}^2/2$ reach a few percent [17, 18]; $\text{Kn} = 0.1$, beyond which slip and non-continuum effects appear) for three fluids.

Fluid (300 K)	$\ell_* = \nu/c_s$	$t_* = \nu/c_s^2$	$t(\text{Ma}=0.3)$	$t(\text{Kn}=0.1)$	$t(\text{Kn}=1)$	ΔT at $\text{Ma} \ 0.3/1$	$ \omega _{\text{max}} = c_s^2/\nu$
Water	0.67 nm	0.44 ps	5–7 ps	9 ps	90 fs	48 K / 540 K	$2.2 \times 10^{12} \text{ s}^{-1}$
Air	45 nm	0.13 ns	1.5–1.8 ns	30 ns	0.3 ns	11 K / 120 K	$7.5 \times 10^9 \text{ s}^{-1}$
Helium	121 nm	0.12 ns	1.3–1.6 ns	27 ns	0.27 ns	18 K / 195 K	$8.3 \times 10^9 \text{ s}^{-1}$

Table 4: Validity limits from Proposition 5.1. Inputs at 300 K (ν , c_s , c_p , λ): water $10^{-6} \text{ m}^2/\text{s}$, 1500 m/s, 4184 J/(kg K), 0.3 nm; air 1.56×10^{-5} , 343, 1005, 68 nm; helium 1.22×10^{-4} , 1007, 5193, 180 nm. Time ranges span the unit-free estimate $\nu/(\text{Ma} c_s)^2$ and the $\ell_0 = 1$ cm value. The last column is the local vorticity bound of §5.7.

5.3 The Mach trajectory for a 1 cm water vortex

For liquid water at reference ambient conditions ($T_0 = 300$ K, $c_s = 1500$ m/s, $\nu = 10^{-6}$ m²/s), initialized with vortex radius $\ell_0 = 1$ cm and reference viscous velocity $u_0 = \nu/\ell_0 = 10^{-4}$ m/s, the collapse profile $u_\theta \sim u_0 \tau^{-1/2-h}$ with $h = 1/200$ yields the trajectory in Table 5; the constant-density idealization is used, by convention, while $\text{Ma} \lesssim 0.3$. The time column is $t = T\tau$ with $T = \ell_0^2/\nu = 100$ s.

Time $t = T\tau$ (s)	Velocity $ u $ (m/s)	Mach Ma	Core Radius ℓ_r (m)	Physical Regime
1.00×10^2	1.00×10^{-4}	6.67×10^{-8}	1.00×10^{-2}	Incompressible Continuum
1.00×10^{-4}	1.07×10^{-1}	7.14×10^{-5}	1.00×10^{-5}	Incompressible Continuum
1.00×10^{-10}	1.15×10^2	7.65×10^{-2}	1.00×10^{-8}	Laminar High-Speed Core
6.69×10^{-12}	4.50×10^2	0.30	2.59×10^{-9}	Incompressibility Limit Breached
6.16×10^{-13}	1.50×10^3	1.00	7.85×10^{-10}	Transonic Shock Wave Formation
9.00×10^{-14}	3.96×10^3	2.64	3.00×10^{-10}	Molecular Mean Free Path Limit ($\text{Kn} = 1$)
1.00×10^{-16}	1.23×10^5	82.0	1.00×10^{-11}	Sub-Molecular / Kinetic Regime
1.00×10^{-23}	4.22×10^8	2.81×10^5	3.16×10^{-15}	Formally Superluminal ($ u > c$) — model has no meaning

Table 5: Trajectory of the OpenAI collapsing vortex core in water ($c_s = 1500$ m/s, $T_0 = 300$ K), with the time column corrected to physical seconds $t = T\tau$, $T = \ell_0^2/\nu = 100$ s. The incompressible model self-invalidates at $t \approx 6.69 \times 10^{-12}$ s (≈ 6.7 picoseconds before the mathematical singularity), and crosses the sonic barrier at $t \approx 6.16 \times 10^{-13}$ s. Velocity, Mach and radius columns are functions of the dimensionless τ only and are unaffected by the time-unit correction.

5.4 Acoustic radiation is not an independent check

Lighthill’s acoustic analogy [15, 16] gives, for a compact *subsonic* source of velocity U and size L , a radiated power $P_{\text{ac}} \sim \rho_0 U^8 L^2 / c_s^5$. Comparing the energy this would radiate over the remaining time t with the kinetic energy of the core, $E_{\text{core}} \sim \rho_0 U^2 \ell_r^2 \ell_z$, and using Eq. (2) for U , ℓ_r and $\ell_z \simeq \ell_r t^{-h}$, one finds

$$\frac{P_{\text{ac}} t}{E_{\text{core}}} \sim \left(\frac{U}{c_s} \right)^5 = \text{Ma}^5 \quad (4)$$

up to $O(1)$ constants and factors t^{-5h} . An earlier draft of this audit presented the time at which this ratio reaches unity as an “independent radiative energy-budget check” that “lands within an order of magnitude” of the Mach limit. It is not independent: it is the Mach criterion again, which is why it agreed with it. The correct reading is simply that the flow ceases to be an acoustically compact, subsonic source at $\text{Ma} \sim 1$ — exactly where Eq. (3) says the incompressible model has ended.

5.5 Viscous heating: the temperature the model cannot see

Incompressible Navier–Stokes carries no temperature equation (§4), so the heating is an external estimate; Proposition 5.1(3) makes it a sharp one. Because the core radius is the viscous length $\sqrt{\nu t}$, the heat released by viscous dissipation over the remaining time equals the local kinetic energy: $\Delta T \simeq u^2/c_p$. For water this is 48 K at $\text{Ma} = 0.3$ and 540 K at $\text{Ma} = 1$; the boiling point ($\Delta T = 73$ K) is reached at $\text{Ma} \approx 0.37$, about 3–4.5 ps before blow-up. Earlier drafts of this audit spoke of “plasma temperatures”; a rise of 10^4 K would require $\text{Ma} \approx 4$, far past the point where the model has ended, and the $\tau^{-1.010}$ law of Table 3 must not be extrapolated there. Two caveats: the estimate assumes the heat stays in the core over the collapse time, which holds when the thermal diffusion length $\sqrt{\alpha t} = \text{Pr}^{-1/2} \ell_r$ does not exceed ℓ_r — true for water ($\text{Pr} \approx 7$), marginal for air ($\text{Pr} \approx 0.7$), where ΔT is an upper estimate; and the model’s own energy balance is exact and its dissipation is positive, so *no thermodynamic law is violated* — what fails is the constitutive assumption of a decoupled, constant temperature.

5.6 Liquids: cavitation comes first

For a liquid at ambient pressure the first constitutive assumption to fail is neither incompressibility nor the continuum but the absence of a vapour phase. The pressure at the centre of a vortex core lies below the ambient pressure by roughly ρu^2 (Bernoulli), so the local pressure reaches the vapour pressure when $\frac{1}{2}\rho u^2 \approx p_\infty - p_v \approx 0.1$ MPa, i.e. at $u \approx 14$ m/s ($\text{Ma} \approx 0.01$). By Eq. (2) this happens $t \simeq \nu/u^2 \approx 5$ ns before blow-up, when the core radius is ≈ 70 nm — three orders of magnitude earlier in time than the Mach limit, in a core still a hundred molecules wide. A nanometre-scale core that lives for picoseconds may not nucleate at p_v : the measured tensile strength of clean water is ~ 30 MPa, and ~ 140 MPa has been reached in mineral inclusions [30, 31]. Those tensions correspond to $u \approx 245$ – 530 m/s, $t \approx 17$ – 4 ps and $\ell_r \approx 4$ – 2 nm — before or together with $\text{Ma} = 0.3$. Whether the liquid cavitates at 5 ns or holds tension to 5 ps, it stops being the fluid the incompressible model describes before, or at, the limits of Proposition 5.1. For gases there is no such limit and Eq. (3) stands alone.

5.7 A single admissibility condition

The limits of Proposition 5.1 combine into one local condition. A vortex of velocity U and size L has $|\omega| \sim U/L$; validity requires $U \lesssim c_s$ (Mach) and $L \gtrsim \nu/c_s$ (Knudsen, through ℓ_*), hence

$$|\omega| \lesssim \frac{c_s^2}{\nu}, \quad \text{equivalently} \quad \varepsilon = \nu|\omega|^2 \lesssim \frac{c_s^4}{\nu}, \quad (5)$$

about $2.2 \times 10^{12} \text{ s}^{-1}$ ($\varepsilon \lesssim 5 \times 10^{18} \text{ W/kg}$) in water and $7.5 \times 10^9 \text{ s}^{-1}$ in air. This is the honest form of the “bounded enstrophy” admissibility axiom proposed in earlier drafts of this project and in its companion work stream on entropy conditions: local, dimensionally derived and material-specific, rather than a global integral bound — which scales with the volume of fluid considered, and for which no derivation was ever given. It also turns the admissibility statement from a definition into a theorem. By Beale–Kato–Majda [8], which holds for Navier–Stokes as well as Euler, a solution with $\int_0^T \|\omega(t)\|_{L^\infty} dt < \infty$ is smooth on $[0, T]$; a solution that satisfies Eq. (5) on $[0, T)$ therefore does not blow up at T , and any blow-up must first leave the admissible set — in this construction at ℓ_* , or in a liquid at the cavitation scale. The condition says nothing about whether real flows approach the bound; it says where the model must be replaced by another.

5.8 What laboratory vortices do instead

Laboratory experiments on interacting vortex filaments in water [19] show reconnection rather than self-similar collapse: core lines approach, pinch off and rearrange, shedding energy into waves and small-scale motion. That is a different configuration from the single swirling column of the construction, and it does not contradict the theorem; it is a reminder that real vortices find other routes to dissipation long before molecular scales. Likewise, extreme dissipation events in direct numerical simulations and high-Reynolds-number experiments [20, 21] are intense but finite and resolved at the Kolmogorov scale η , which in every measured flow exceeds the molecular length by orders of magnitude. The Beale–Kato–Majda criterion [8] is the mathematical statement that this finiteness is what prevents blow-up; it is not itself a statement about molecules.

6 Backward-Specified Forcing and the Limits of MMS

Physics problems are usually posed forward: specify a force, then compute how the fluid responds. This proof’s logical structure runs the specification backward. The authors:

1. **Stipulate** the goal: a singularity at $t = 1$.

2. **Engineer** a collapsing vortex column whose velocity diverges (Section 4 of [2]).
3. **Compute** the raw residual, which is singular ($f_{\text{raw}} \rightarrow \infty$).
4. **Construct** oscillatory wave packets (Sections 6–9) whose internal Reynolds stresses cancel the singular residual.
5. **Declare** the remaining smooth remainder as the external force $f(x, t)$ (Section 10).

This is a completely legitimate way to build an existence proof — fixing the endpoint and solving backward for compatible data is standard technique — but it has a specific consequence worth flagging: because (u, f) are co-designed as a single package, f cannot be given an independent physical origin (a stirring mechanism, a boundary forcing, a body force from an external field) without already knowing the singular solution it is meant to produce. The paper is transparent about this itself: “An exponentially small external force seeds each pulse; the background shear supplies its subsequent growth.” The force *seeds* — it does not *drive*.

It is, in that sense, the mathematical equivalent of shooting an arrow into a wall and painting a bullseye around it: a legitimate demonstration that the wall *can* be hit, not a claim about where arrows normally land.

7 The Unforced Euler Singularity: Coherent Structure Below Any Physical Cutoff

For the unforced Euler equations the singularity must come from the initial datum alone. The construction [4] builds $u_0 \in C_{c,\sigma}^\infty(\mathbb{R}^3)$ by a stagewise iteration: at each stage a localized oscillatory packet of higher frequency is added to the flow built so far, the packet is amplified by the shear of that parent flow, and its averaged momentum flux feeds the next stage; amplitudes, frequencies, localization lengths and time intervals are chosen so that the sum converges in every fixed Sobolev norm while $\int_0^{T_*} \|\omega(t)\|_{L^\infty} dt$ diverges at a finite T_* ([4, §§2–5]).

Because u_0 is C^∞ , its energy below any molecular length λ is negligible; the physical issue is not energy but *coherence*. The mechanism requires the packets, down to arbitrarily fine scales, to be present in the datum with prescribed phases so that the shear of the coarser stages can amplify them in the right order. In a real fluid the Euler equations are the $\nu \rightarrow 0$ idealization of Navier–Stokes, so the physics missing from the Euler construction is viscosity itself, which damps a packet of wavenumber k on the time $1/(\nu k^2)$ — together with thermal fluctuations, which set an irreducible noise floor near the scale $\ell_{\text{th}} = k_B T / (\rho \nu^2)$ (a few picometres in water, i.e. already below λ , so the molecular length is the operative cutoff [29]). Whether a viscous, noisy fluid can reproduce the first few stages of the cascade before the fine scales are washed out is a computable question (§10), not a slogan. Earlier drafts of this audit described the fine scales as “sub-Planckian”; the Planck length is irrelevant here — the cutoff is molecular.

8 Properties of the Specific Construction, and What Was Removed

Table 6 collects properties of the particular solution built. They are properties of that solution, not defects in the proof, and none of them is a “vulnerability” in the sense earlier drafts of this audit used the word.

8.1 Numerical material removed from this version

Earlier drafts of this audit included Taylor–Green-vortex runs from an in-house pseudo-spectral solver, a “Banach contraction-rate certificate”, and a figure contrasting a JHTDB isotropic-turbulence spectrum with an “OpenAI pre-singularity spectrum”. We have removed them. The

Property	OpenAI Formalization	Reading
1. Backward-specified residual	<code>force := SpacetimeGluingsmoothExtension 1 (tracedResidual u p L) (CandidateFromLimits.lean)</code>	The force is the smooth remainder of a co-designed (u, p) (§6); initial data is $u_0 \equiv 0$ for the forced case.
2. One explicit datum	<code>$\exists u_0, \text{ContDiff } \mathbb{R} \propto u_0$</code>	A single point of $C_{c,\sigma}^\infty$. Its codimension — the number of unstable directions of the linearized self-similar profile beyond the symmetry modes — is the rigorous measure of “non-generic” and is not established by this audit (§10).
3. Axisymmetric swirl	<code>AxisymmetricFields.lean</code>	A 2D similarity profile with swirl; it need not represent generic three-dimensional turbulence.
4. No dissipative floor	Corrections summed “with shrinking cutoffs” down to $\ell_r \rightarrow 0$	There is no Kolmogorov scale η ; a physical cut-off enforcing Eq. (5) would prevent blow-up by Beale–Kato–Majda.
5. Engineered life-time	<code>SmoothCutoffs.lean</code> time switch	Forcing is compactly supported in time; the lifespan $T^* = 1$ is a choice, not a feature of the equations.

Table 6: Properties of the constructed solution, with the Lean locations that encode them.

TGV numbers were not reported with resolution and normalization sufficient for independent reproduction; the certificate checked the internal consistency of a software pipeline, not any property of the construction; and the “OpenAI spectrum” was a schematic Gaussian spike drawn by hand, not a computed spectrum — no numerical realization of the 166-page construction exists, in this project or, to our knowledge, elsewhere. What a genuine numerical companion should contain is listed in §10: the cutoff law, the codimension count and the fluctuating-hydrodynamics questions. The scripts distributed with this repository are exponent, unit and consistency checks, and are labelled as such.

9 Discussion

The OpenAI system’s achievement is a genuine milestone in automated reasoning. It exploited the analytical gap between C^∞ and real-analytic functions, using Gevrey-class cutoffs to construct a mathematically flawless singularity that the Lean 4 kernel rigorously certifies. It did not cheat; it solved the mathematical problem as posed.

Five observations, taken together, show that the constructed singularity exists within the abstract mathematical continuum $\mathbb{R}^3$ without describing anything a real, physical fluid does: (1) intensive local energy density and enstrophy diverge ($\sim \tau^{-1.010}$, $\sim \tau^{-0.515}$) while the global energy vanishes; (2) compressibility, rarefaction and viscous heating all become order-one at a single scale $\ell_* = \nu/c_s$ — a few picoseconds before blow-up in water, nanoseconds in air, essentially independently of the initial vortex size — and in a liquid cavitation intervenes three decades earlier still; (3) the moment-matching system’s apparent ill-conditioning is a non-dimensionalization artifact, not physical fragility; (4) the force is backward-specified and has no independent physical origin; and (5) the Euler datum needs coherent structure below the molecular length, where the missing physics is viscosity.

This is not, we want to stress, a novel or contested position among fluid dynamicists: it is close to the median reaction from the community discussion summarized in §10, and it echoes a long-standing point about continuum models generally — Ohm’s law, ideal-gas equations of state, and other workhorse relations all have known breakdown regimes without this being considered a mark against their mathematical content. Three implications follow:

1. **For mathematics:** The proof is definitive. Alternatives C and D are resolved as statements

about the continuum incompressible model.

2. **For physics:** This particular singularity requires the fluid to remain a continuum, incompressible, isothermal medium down to ℓ_* — a few molecular diameters in water — and, in a liquid, to sustain tensions of order 10^2 MPa for picoseconds. It is not an observable. What *is* observable is the question of how a real fluid leaves the model: cavitation, compressibility, rarefaction and heating, in an order that Proposition 5.1 makes explicit.
3. **For AI-assisted mathematics:** A kernel-checked proof of a physically motivated equation is not automatically a physical result, and the gap between the two is worth making explicit — and, where possible, checkable — as part of the proof pipeline (§10.3).

10 Outlook: Repositioning, Community Directions, and Physically-Grounded AI

10.1 Mathematics at Its Limit, Not a Claim About Physics

Section 1 already previewed the position we think the evidence supports, and which was arrived at independently by much of the fluid-dynamics community when an early version of this audit was first posted for feedback on `r/FluidMechanics` by one of the present authors [6]: the Millennium Prize problems, Navier–Stokes existence and smoothness included, were set as precise tests of mathematical technique against a specific, well-defined continuum model — not as referenda on whether that model perfectly describes real fluids. As one commenter (`Leodip`) put it, “the point of the millennium prize problem was never to figure out anything with regards to real world fluids... finite-time blowup is totally unphysical either way and just a result of the N-S model.” Another (`mrbiguri`) added that a blow-up result “suggests that it’s not fully descriptive of real world physics,” and a third (`Dihedralman`) compared this to other well-known breakdowns of continuum models, noting simply that “Ohm’s law breaks down... Navier Stokes has always been a useful approximation and not fully physical.”

A fourth exchange, prompted by a question (`brokebeany`) about whether the incompressibility breakdown identified in §5 invalidates the OpenAI proof, produced what we think is the cleanest statement of the right frame (`Fluid_Substance5132`): “In the real world[,] the incompressible approximation fails to capture what actually happens. Within the (non physical) incompressible world[,] the o[pen]ai proof remains valid.”

We agree with that statement without qualification. Everything in §2–§7 should be read accordingly: not as claims that the Lean proof is wrong, but as a map of exactly where and how far this specific constructed solution departs from the regime the incompressible model is normally trusted to describe. That map is, we think, the useful residue of this kind of audit, and several commenters independently proposed uses for it that go well beyond simply flagging unphysicality, summarized next.

10.2 Research Directions from the Community Response

Four concrete directions recurred in that discussion and align with work streams already underway in this project’s companion directory, `05_Community_Research_Directions/`, which credits the same discussion:

1. **An admissibility, or entropy, condition for 3D Navier–Stokes.** `cowgod42` proposed that 3D Navier–Stokes may need an augmented admissibility condition, analogous to the Lax entropy condition used to select the physically correct weak solution of the inviscid Burgers equation, to filter out unphysical weak solutions such as this one; they noted the blow-up result tells us “the ‘correct’ equations of fluids should not allow this to happen...”

we need to change the equations, or augment them in some way.” `tlmBot` independently suggested the problem itself may need restatement so that it pertains only to physical fluids. This direction is being developed in `WorkStream1_EntropyCondition`.

2. **The pre-singularity trajectory as a turbulence-modeling resource.** `babainottawa` and `Cute_Presence_2861` both noted that even though the endpoint is unphysical, the approach to it — before the Mach or Knudsen thresholds in Table 5 are crossed — may still carry information relevant to subgrid-scale closure modeling (LES/RANS). This direction is being developed in `WorkStream3_Turbulence`.
3. **Deriving, rather than assuming, the thermal response.** `vibe0009` correctly pointed out that incompressible Navier–Stokes has no energy equation, so any thermal-shock argument built on top of it (including ours in §4) is an external estimate, not a derived consequence. A genuine energy/temperature-coupled extension, or a cavitation/boiling model, is needed before the thermal argument can be sharpened past an order-of-magnitude check. This direction is being developed in `WorkStream2_ThermalEquation` and `WorkStream4_MechanicalCavitation`.
4. **Teaching proof-search AI systems basic thermodynamics.** In a follow-up post summarizing the initial round of feedback, the audit’s authors distilled the most actionable community suggestion as: “Figuring out how to teach AI proof assistants basic thermodynamics so they catch these edge cases.” That suggestion is the direct motivation for §10.3.

None of these four directions is resolved by this paper; we record them as open invitations, consistent with the standing call for contributions in `05_Community_Research_Directions/README.md`.

10.3 Toward Physically-Grounded AI Theorem Proving

Motivated directly by the fourth point above, we propose a three-pillar architecture for future AI-assisted formalization of physically motivated equations, restating an earlier internal draft manifesto (`TAO_NEUROSymbolic_Scientific_AI_Manifesto.md`) in a form scoped to what is realistically buildable rather than what is aspirational:

1. **Neural proposal.** An LLM-based multi-agent system, as in OpenAI’s own architecture, proposes candidate constructions, tactics, and similarity profiles.
2. **Symbolic verification.** A Lean 4 (or equivalent) kernel checks each candidate for syntactic and logical correctness — exactly as today, with zero `sorry` and zero unauthorized axioms. This pillar alone is what makes a result a *theorem*; nothing below changes that.
3. **Physical admissibility gate.** A third, independent checker evaluates the *same* construction against an explicit, falsifiable list of physical admissibility predicates before it is reported as *physically* relevant (not before it is accepted as a mathematical theorem). For incompressible Navier–Stokes the list is short and dimensionally derived (§5.7): the local vorticity bound $|\omega| \lesssim c_s^2/\nu$ of Eq. (5), which encodes $\text{Ma} \lesssim 1$ and $\text{Kn} \lesssim 1$ at once; for liquids, a core pressure deficit ρu^2 below the tensile strength; and, where an energy equation is present, an Eckert number $u^2/(c_p \Delta T)$ that stays small. Each predicate is a scaling estimate with $O(1)$ prefactors, and should be reported as such.

A rational-arithmetic reflection bridge (`openai_poc_pi_verifier.py`) already implements a narrow version of pillar 3’s Mach/Knudsen checks in Lean, using exact rational arithmetic evaluated via `#eval`; it is a reasonable starting point for the broader gate above, though it does not yet reach into the OpenAI proof objects themselves.

We want to be explicit about the current state of this project’s own attempt at pillar 3: the `TopologicalCensorship.lean` module in this repository proves elementary boundedness and monotonicity lemmas about an effective-wavenumber formula and a dissipation-rate/enstrophy inequality in isolation. As of this writing, *none* of those lemmas are connected, inside Lean, to OpenAI’s actual `NavierStokes/Euler` definitions. Presenting them as already constraining the OpenAI construction would repeat exactly the kind of overclaiming earlier drafts of this project were rightly criticized for. Wiring a genuine physical-admissibility gate to the actual proof objects — not to a free-standing toy formula — is, in our view, the concrete next engineering step, and one we have not yet completed.

10.4 A Hypothesis on Cutoff Regularizations: Testable Questions, Not a Metric

Finally, and separately from the buildable architecture above, we state one further-out idea explicitly as a **hypothesis for future investigation**. Earlier drafts of this project proposed a “dual-scale” effective wavenumber saturating at large $|k|$,

$$k_{\text{eff}}(k) = \min \left(|k|, \frac{1}{\alpha' |k|} \right) \quad (k \neq 0), \quad k_{\text{eff}} \leq 1/\sqrt{\alpha'}, \quad (6)$$

with $\alpha' = \ell_{\min}^2$, and motivated it by string-theoretic T-duality and Trans-Planckian censorship [33]. The inequality is true and Lean-checkable (`dual_scale_wavenumber_bounded` in this repository), but it is an algebraic fact about a formula, and the string-theoretic motivation adds nothing: stripped of that language, a regularization of this kind is a member of the Leray- α / Bessel-filter family $\bar{u} = (1 - \alpha' \Delta)^{-2} u$, whose global regularity is known [27, 28]. The draft that asserted a substantive physical link, via a $K3 \times T^2$ compactification, was reviewed and its physical claim withdrawn (its central Lean theorem ended in a bare `True`, and its vorticity equation lacked a term); it now carries a withdrawal notice in this repository, and the companion note [5] records the corrections.

What survives is a hypothesis with a definite test. Drive the regularized model with OpenAI’s force, or a truncation of it. If the collapse is arrested when $\ell_r \sim \sqrt{\alpha'}$, then by Eq. (2) the peak velocity should scale as $u_{\max} \sim \nu/\sqrt{\alpha'}$ and the arrest time as $t_c \sim \alpha'/\nu$; setting $\sqrt{\alpha'} = \ell_*$ gives $u_{\max} \sim c_s$, consistent with Proposition 5.1. A positive answer would turn the regularization into a predictive statement — “a molecular cutoff at ℓ bounds the peak velocity at ν/ℓ ” — that kinetic or molecular-dynamics simulations can check. A negative answer (the pulses de-cohering before ℓ_r reaches $\sqrt{\alpha'}$) would be equally informative. Two companion questions belong with it: the codimension of the blow-up, i.e. the number of unstable modes of the linearized self-similar profile beyond those generated by symmetries, which is the rigorous version of “fine-tuned” and has been answered for related Euler and Boussinesq profiles by computer-assisted and neural-network methods [24, 25]; and the fate of the exponentially small seeding force under Landau–Lifshitz thermal noise [29]. We claim none of these results; we claim only that they are the right questions.

11 Conclusion

OpenAI’s formalization is a genuine, kernel-verified mathematical achievement, and Alternatives C and D are, as mathematical statements about the continuum incompressible model, resolved by it. Nothing in this paper disputes that, and nothing in it could: the Lean kernel already settled the question of syntactic correctness before we started.

What this paper adds is a physical reading, corrected in several places from earlier drafts of this audit: a scaling table for locally divergent quantities that respects the Prize’s global energy bound; a unit-free validity scale $\ell_* = \nu/c_s$ at which compressibility, rarefaction and heating

enter together, with the earlier fluid-specific timelines recovered as special cases; the observation that in liquids cavitation comes first; a single local admissibility condition $|\omega| \lesssim c_s^2/\nu$ that Beale–Kato–Majda turns into a theorem; a demonstration that the moment-matching system’s apparent ill-conditioning is a non-dimensionalization artifact, and that the acoustic “check” of earlier drafts was the Mach criterion in disguise; an honest accounting of the backward-specified force; and a description of the Euler datum’s dependence on coherent structure below the molecular length. None of these observations makes the mathematics wrong, and none of them, on their own, proves that real turbulent fluids cannot develop extreme, short-lived, localized events resembling a scaled-down piece of this construction — that remains a genuinely open empirical and theoretical question, not one this audit resolves. What they do establish is that *this particular* construction is not a description of anything a real fluid does; it is a demonstration of what the continuum model, pushed to the exact edge of its own internal consistency, is mathematically capable of.

The more useful outcome of this exercise, in our view, is not the verdict but the map: a concrete list of where the physical assumptions break down, a set of research directions raised independently by the community that received this audit (§10), a proposed — and only partially built — architecture for keeping future AI-generated proofs of physical PDEs honestly labeled as mathematical or physical results, and one explicitly speculative hypothesis for anyone who wants to push the dual-scale idea further than we have. The AI solved the mathematician’s problem. Whether, and how, that result connects to the physicist’s problem is still an open question, and, we think, a more interesting one.

References

- [1] C.L. Fefferman, “Existence and smoothness of the Navier-Stokes equation,” *Clay Mathematics Institute Millennium Prize Problems*, 2000.
- [2] OpenAI, “Finite time blowup for Navier-Stokes and Euler equations,” Lean 4 Formalization, github.com/openai/NavierStokesAndEuler, 2026.
- [3] OpenAI, “On the Navier–Stokes Millennium Prize Problem,” September 8, 2026, <https://openai.com/index/navier-stokes-solution/>.
- [4] OpenAI, “Finite time blowup for the Euler equation,” preprint and Lean 4 formalization, github.com/openai/NavierStokesAndEuler, 2026.
- [5] X. Callens, “Where the continuum ends: the OpenAI Navier–Stokes blow-up as a probe of model validity, two-scale closure, and formalized physics,” version 2, this repository ([paper/where_the_continuum_ends.tex](https://github.com/XCallens/where_the_continuum_ends)), September 2026.
- [6] r/FluidMechanics, “The fluid dynamics behind OpenAI’s Navier-Stokes proof: Why the fluid would vaporize before the singularity” and follow-up discussion, thread by u/OrganizationTop9026, September 2026, <https://www.reddit.com/r/FluidMechanics/comments/1wekxy3/>.
- [7] L. Escauriaza, G. Seregin, V. Šverák, “ $L_{3,\infty}$ -solutions of Navier-Stokes equations and backward uniqueness,” *Russian Math. Surveys* **58**(2), 2003.
- [8] J. T. Beale, T. Kato, A. Majda, “Remarks on the breakdown of smooth solutions for the 3-D Euler equations,” *Comm. Math. Phys.* **94**, 61–66, 1984.
- [9] R. Penrose, “Gravitational collapse: The role of general relativity,” *Riv. Nuovo Cimento* **1**, 252–276, 1969.

- [10] T. Tao, “Finite time blowup for an averaged three-dimensional Navier-Stokes equation,” *J. Amer. Math. Soc.* **29**(3), 601–674, 2016.
- [11] L. Caffarelli, R. Kohn, L. Nirenberg, “Partial regularity of suitable weak solutions of the Navier-Stokes equations,” *Comm. Pure Appl. Math.* **35**(6), 771–831, 1982.
- [12] J. Leray, “Sur le mouvement d’un liquide visqueux emplissant l’espace,” *Acta Math.* **63**, 193–248, 1934.
- [13] P. Constantin, C. Foias, *Navier-Stokes Equations*, University of Chicago Press, 1988.
- [14] A. N. Kolmogorov, “The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers,” *Dokl. Akad. Nauk SSSR* **30**, 299–303, 1941.
- [15] M. J. Lighthill, “On sound generated aerodynamically. I. General theory,” *Proc. R. Soc. Lond. A* **211**(1107), 564–587, 1952.
- [16] I. Proudman, “The generation of noise by isotropic turbulence,” *Proc. R. Soc. Lond. A* **214**(1116), 119–132, 1952.
- [17] L. D. Landau, E. M. Lifshitz, *Fluid Mechanics* (Volume 6 of Course of Theoretical Physics, 2nd ed.), Pergamon Press, 1987.
- [18] G. K. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press, 1967.
- [19] D. Kleckner, W. T. M. Irvine, “Creation and dynamics of knotted vortices,” *Nature Physics* **9**(4), 253–258, 2013.
- [20] J. Schumacher, J. D. Scheel, D. Krasnov, D. A. Donzis, V. Yakhot, K. R. Sreenivasan, “Small-scale universality in fluid turbulence,” *Proc. Natl. Acad. Sci. USA* **111**(30), 10961–10965, 2014.
- [21] D. A. Donzis, P. K. Yeung, K. R. Sreenivasan, “Dissipation and enstrophy in isotropic turbulence: resolution effects and scaling in direct numerical simulations,” *Phys. Fluids* **20**, 045108, 2008.
- [22] G. Luo, T. Y. Hou, “Toward the finite-time blowup of the 3D axisymmetric Euler equations: a numerical investigation,” *Multiscale Model. Simul.* **12**(4), 1722–1776, 2014.
- [23] T. M. Elgindi, “Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$,” *Ann. of Math.* **194**(3), 647–727, 2021.
- [24] J. Chen, T. Y. Hou, “Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data,” arXiv:2210.07191, 2022.
- [25] Y. Wang et al., “Discovery of unstable singularities,” arXiv:2509.14185, 2025.
- [26] D. Albritton, E. Brué, M. Colombo, “Non-uniqueness of Leray solutions of the forced Navier–Stokes equations,” *Ann. of Math.* **196**(1), 415–455, 2022.
- [27] C. Foias, D. D. Holm, E. S. Titi, “The Navier–Stokes-alpha model of fluid turbulence,” *Physica D* **152–153**, 505–519, 2001.
- [28] A. Cheskidov, D. D. Holm, E. Olson, E. S. Titi, “On a Leray- α model of turbulence,” *Proc. R. Soc. A* **461**, 629–649, 2005.
- [29] D. Bandak, N. Goldenfeld, A. A. Mailybaev, G. Eyink, “Dissipation-range fluid turbulence and thermal noise,” *Phys. Rev. E* **105**, 065113, 2022.

- [30] Q. Zheng, D. J. Durben, G. H. Wolf, C. A. Angell, “Liquids at large negative pressures: water at the homogeneous nucleation limit,” *Science* **254**, 829–832, 1991.
- [31] F. Caupin, E. Herbert, “Cavitation in water: a review,” *C. R. Physique* **7**, 1000–1017, 2006.
- [32] T. Buckmaster, V. Vicol, “Nonuniqueness of weak solutions to the Navier-Stokes equation,” *Ann. of Math.* **189**(1), 101–144, 2019.
- [33] A. Bedroya, C. Vafa, “Trans-Planckian censorship and the swampland,” *J. High Energ. Phys.* **2020**(9), 123, 2020.